
This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Camera
Manufacturer: AViPAS
Firmware Version: N/A
Model(s): AV-2020

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.14.0000-b007	2.13.0

Version History

Module Version	Date	Notes
1_0_0_0	11/2/2021	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' to '7'.
Example: `InterfaceName.DeviceID = '1'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='AV-2020')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='AV-2020')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 52381, Model='AV-2020')</code>

Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AutoExposure	Value 'Full Auto' 'Iris Priority'	Value 'Manual' 'Bright'	Value 'Shutter Priority'
# AutoExposure example InterfaceName.Set('AutoExposure', 'Full Auto')			
Command AutoFocus	Value 'On'	Value 'Off'	
# AutoFocus example InterfaceName.Set('AutoFocus', 'On')			
Command Backlight	Value 'On'	Value 'Off'	
# Backlight example InterfaceName.Set('Backlight', 'On')			
Command Focus	Value 'Far'	Value 'Near'	Value 'Stop'
Qualifier Key 'Speed'	Qualifier Value 0 – 7		
# Focus example InterfaceName.Set('Focus', 'Far', {'Speed': 7})			
Command Gain	Value 'Up'	Value 'Down'	Value 'Reset'
# Gain example InterfaceName.Set('Gain', 'Up')			
Command Iris	Value 'Up'	Value 'Down'	Value 'Reset'
# Iris example InterfaceName.Set('Iris', 'Up')			
Command PanTilt	Value 'Up' 'Right' 'Down Left' 'Home'	Value 'Down' 'Up Left' 'Down Right' 'Reset'	Value 'Left' 'Up Right' 'Stop'
Qualifier Key 'Pan Speed'	Qualifier Value 1 – 24		
Qualifier Key 'Tilt Speed'	Qualifier Value 1 – 20		
# PanTilt example InterfaceName.Set('PanTilt', 'Up', {'Pan Speed': 24, 'Tilt Speed': 20})			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			

**Global Scriptor Module
Communication Sheet**

Command Preset	Value 1 – 128		
Qualifier Key 'Action'	Qualifier Value 'Save'	Qualifier Value 'Recall'	Qualifier Value 'Reset'
# Preset example InterfaceName.Set('Preset', 128, {'Action': 'Save'})			
Command Shutter	Value 'Up'	Value 'Down'	Value 'Reset'
# Shutter example InterfaceName.Set('Shutter', 'Up')			
Command Zoom	Value 'Tele'	Value 'Wide'	Value 'Stop'
Qualifier Key 'Speed'	Qualifier Value 0 – 7		
# Zoom example InterfaceName.Set('Zoom', 'Tele', {'Speed': 7})			

Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})  
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})  
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},  
FeedbackHandler)
```

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)  
Value = InterfaceName.ReadStatus(Command)  
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)  
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AutoExposure	Value 'Full Auto' 'Iris Priority'	Value 'Manual' 'Bright'	Value 'Shutter Priority'
# AutoExposure example InterfaceName.Update('AutoExposure') Value = InterfaceName.ReadStatus('AutoExposure') InterfaceName.SubscribeStatus('AutoExposure', None, FeedbackHandler)			
Command AutoFocus	Value 'On'	Value 'Off'	
# AutoFocus example InterfaceName.Update('AutoFocus') Value = InterfaceName.ReadStatus('AutoFocus') InterfaceName.SubscribeStatus('AutoFocus', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			

Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AutoExposure	Value 'Full Auto' 'Iris Priority'	Value 'Manual' 'Bright'	Value 'Shutter Priority'
# AutoExposure example InterfaceName.Set('AutoExposure', 'Full Auto')			
Command AutoFocus	Value 'On'	Value 'Off'	
# AutoFocus example InterfaceName.Set('AutoFocus', 'On')			
Command Backlight	Value 'On'	Value 'Off'	
# Backlight example InterfaceName.Set('Backlight', 'On')			
Command Focus	Value 'Far'	Value 'Near'	Value 'Stop'
Qualifier Key 'Speed'	Qualifier Value 0 – 7		
# Focus example InterfaceName.Set('Focus', 'Far', {'Speed': 7})			
Command Gain	Value 'Up'	Value 'Down'	Value 'Reset'
# Gain example InterfaceName.Set('Gain', 'Up')			
Command Iris	Value 'Up'	Value 'Down'	Value 'Reset'
# Iris example InterfaceName.Set('Iris', 'Up')			
Command PanTilt	Value 'Up' 'Right' 'Down Left' 'Home'	Value 'Down' 'Up Left' 'Down Right' 'Reset'	Value 'Left' 'Up Right' 'Stop'
Qualifier Key 'Pan Speed'	Qualifier Value 1 – 24		
Qualifier Key 'Tilt Speed'	Qualifier Value 1 – 20		
# PanTilt example InterfaceName.Set('PanTilt', 'Up', {'Pan Speed': 24, 'Tilt Speed': 20})			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			

**Global Scripter Module
Communication Sheet**

Command Preset	Value 1 – 128		
Qualifier Key 'Action'	Qualifier Value 'Save'	Qualifier Value 'Recall'	Qualifier Value 'Reset'
# Preset example InterfaceName.Set('Preset', 128, {'Action': 'Save'})			
Command Shutter	Value 'Up'	Value 'Down'	Value 'Reset'
# Shutter example InterfaceName.Set('Shutter', 'Up')			
Command Zoom	Value 'Tele'	Value 'Wide'	Value 'Stop'
Qualifier Key 'Speed'	Qualifier Value 0 – 7		
# Zoom example InterfaceName.Set('Zoom', 'Tele', {'Speed': 7})			

Status Available (Ethernet)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})  
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})  
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},  
FeedbackHandler)
```

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)  
Value = InterfaceName.ReadStatus(Command)  
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)  
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AutoExposure	'Full Auto' 'Iris Priority'	'Manual' 'Bright'	'Shutter Priority'
# AutoExposure example InterfaceName.Update('AutoExposure') Value = InterfaceName.ReadStatus('AutoExposure') InterfaceName.SubscribeStatus('AutoExposure', None, FeedbackHandler)			
Command	Value	Value	
AutoFocus	'On'	'Off'	
# AutoFocus example InterfaceName.Update('AutoFocus') Value = InterfaceName.ReadStatus('AutoFocus') InterfaceName.SubscribeStatus('AutoFocus', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	
Power	'On'	'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			

Cable and Adapter Requirements

(RS-232) Captive Screw to Male Mini-DIN 8-Pin RS-232 Serial Cable

(RS-485) Captive Screw to Male Mini-DIN 8-Pin RS-232 Serial Cable

Notes for the Device

To ensure the functionality of the driver, please set the following:

- Menu → SYSTEM → PROTOCOL → VISCA
- Menu → SYSTEM → VISCA → OVER ALL or OVER COM

Serial communication

Port Type: RS-232, RS-485

Baud Rate: 9600

Data Bits: 8

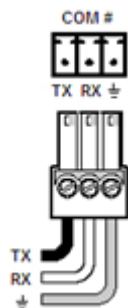
Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram

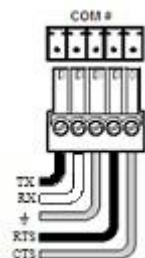
RS-232:



Signal	Main Cable	Pin	Signal
TxD	→	5	RxD
RxD	←	3	TxD
GND		4	GND



RS-485:



Signal	Main Cable	Pin	Signal
TxD	→	8	D-
RxD	→	8	D-
RTS	→	6	D+
CTS	→	6	D+
GND		4	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type:	Ethernet (UDP)
Device Listen Port:	52381
Device Reply Port:	52381
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Listen Port Changeability:	No
Listen Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication.

When using multiple UDP modules within the same configuration each module should have a unique 'Reply on Port' number due to the nature of UDP communication. If you have a driver with a fixed 'Reply on Port' number, only one instance of the module can be used within the configuration.

Notes for the Device

To ensure the functionality of the driver, please set the following:

- Menu → SYSTEM → PROTOCOL → VISCA
- Menu → SYSTEM → VISCA → OVER ALL or OVER IP

Appendix A. Set Commands(Serial, Device ID: 1)

Auto Exposure Bright	\x81\x01\x049\x0D\xff
Auto Exposure Full Auto	\x81\x01\x049\x00\xff
Auto Exposure Iris Priority	\x81\x01\x049\x0B\xff
Auto Exposure Manual	\x81\x01\x049\x03\xff
Auto Exposure Shutter Priority	\x81\x01\x049\x0A\xff
Auto Focus Off	\x81\x01\x048\x03\xff
Auto Focus On	\x81\x01\x048\x02\xff
Backlight Off	\x81\x01\x043\x03\xff
Backlight On	\x81\x01\x043\x02\xff
Focus Far Speed 0	\x81\x01\x04\x08\xff
Focus Far Speed 7	\x81\x01\x04\x08'\xff
Focus Near Speed 0	\x81\x01\x04\x080\xff
Focus Near Speed 7	\x81\x01\x04\x087\xff
Focus Stop Speed 0	\x81\x01\x04\x08\x00\xff
Focus Stop Speed 7	\x81\x01\x04\x08\x00\xff
Gain Down	\x81\x01\x04\x0C\x03\xff
Gain Reset	\x81\x01\x04\x0C\x00\xff
Gain Up	\x81\x01\x04\x0C\x02\xff
Iris Down	\x81\x01\x04\x0B\x03\xff
Iris Reset	\x81\x01\x04\x0B\x00\xff
Iris Up	\x81\x01\x04\x0B\x02\xff
Pan Tilt Down Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\xff
Pan Tilt Down Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x02\xff
Pan Tilt Down Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x02\xff
Pan Tilt Down Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x02\xff
Pan Tilt Down Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x02\xff
Pan Tilt Down Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x02\xff
Pan Tilt Down Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x02\xff
Pan Tilt Down Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x02\xff
Pan Tilt Down Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x02\xff
Pan Tilt Down Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x02\xff
Pan Tilt Down Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x02\xff
Pan Tilt Down Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x02\xff
Pan Tilt Home Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x04\xff
Pan Tilt Home Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x04\xff
Pan Tilt Home Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x04\xff
Pan Tilt Home Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x04\xff
Pan Tilt Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\xff
Pan Tilt Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x03\xff
Pan Tilt Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x03\xff
Pan Tilt Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x03\xff
Pan Tilt Reset Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x05\xff

**Global Scriptor Module
Communication Sheet**

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Pan Tilt Reset Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x05\xff
Pan Tilt Reset Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x05\xff
Pan Tilt Reset Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x05\xff
Pan Tilt Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x02\x03\xff
Pan Tilt Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x03\xff
Pan Tilt Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x03\xff
Pan Tilt Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x03\xff
Pan Tilt Stop Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x03\xff
Pan Tilt Stop Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x03\xff
Pan Tilt Stop Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x03\xff
Pan Tilt Stop Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x03\xff
Pan Tilt Up Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x01\xff
Pan Tilt Up Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x01\xff
Pan Tilt Up Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x01\xff
Pan Tilt Up Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x01\xff
Pan Tilt Up Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x01\xff
Pan Tilt Up Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x01\xff
Pan Tilt Up Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x01\xff
Pan Tilt Up Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x01\xff
Pan Tilt Up Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x01\xff
Pan Tilt Up Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x01\xff
Pan Tilt Up Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x01\xff
Pan Tilt Up Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x01\xff
Power Off	\x81\x01\x04\x00\x03\xff
Power On	\x81\x01\x04\x00\x02\xff
Preset 1 Action Recall	\x81\x01\x04?\x02\x00\xff
Preset 1 Action Reset	\x81\x01\x04?\x00\x00\xff
Preset 1 Action Save	\x81\x01\x04?\x01\x00\xff
Preset 128 Action Recall	\x81\x01\x04?\x02\x7F\xff
Preset 128 Action Reset	\x81\x01\x04?\x00\x7F\xff
Preset 128 Action Save	\x81\x01\x04?\x01\x7F\xff
Shutter Down	\x81\x01\x04\x0A\x03\xff
Shutter Reset	\x81\x01\x04\x0A\x00\xff
Shutter Up	\x81\x01\x04\x0A\x02\xff
Zoom Stop Speed 0	\x81\x01\x04\x07\x00\xff
Zoom Stop Speed 7	\x81\x01\x04\x07\x00\xff
Zoom Tele Speed 0	\x81\x01\x04\x07 \xff
Zoom Tele Speed 7	\x81\x01\x04\x07' \xff
Zoom Wide Speed 0	\x81\x01\x04\x070\xff
Zoom Wide Speed 7	\x81\x01\x04\x077\xff

Appendix B. Update Commands(Serial, Device ID: 1)

Auto Exposure	\x81\x09\x049\xff
Auto Focus	\x81\x09\x048\xff
Power	\x81\x09\x04\x00\xff

Appendix C. Set Commands(Ethernet)

Auto Exposure Full Auto	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x049\x00\xFF
Auto Exposure Iris Priority	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x049\x0B\xFF
Auto Exposure Manual	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x049\x03\xFF
Auto Exposure Shutter Priority	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x049\x0A\xFF
Auto Focus Off	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x048\x03\xFF
Auto Focus On	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x048\x02\xFF
Backlight Off	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x043\x03\xFF
Backlight On	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x043\x02\xFF
Focus Far Speed 0	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x08 \xFF
Focus Far Speed 7	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x08 '\xFF
Focus Near Speed 0	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x080\xFF
Focus Near Speed 7	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x087\xFF
Focus Stop Speed 0	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x08\x00\xFF
Focus Stop Speed 7	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x08\x00\xFF
Gain Down	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0C\x03\xFF
Gain Reset	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0C\x00\xFF
Gain Up	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0C\x02\xFF
Iris Down	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0B\x03\xFF
Iris Reset	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0B\x00\xFF
Iris Up	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0B\x02\xFF
Pan Tilt Down Left Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x02\xFF
Pan Tilt Down Left Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x14\x01\x02\xFF
Pan Tilt Down Left Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x01\x02\xFF
Pan Tilt Down Left Pan Speed 24 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x14\x01\x02\xFF
Pan Tilt Down Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x03\x02\xFF
Pan Tilt Down Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x14\x03\x02\xFF
Pan Tilt Down Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x03\x02\xFF
Pan Tilt Down Pan Speed 24 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x14\x03\x02\xFF
Pan Tilt Down Right Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x02\x02\xFF
Pan Tilt Down Right Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x14\x02\x02\xFF
Pan Tilt Down Right Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x02\x02\xFF
Pan Tilt Down Right Pan Speed 24 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x14\x02\x02\xFF

**Global Scriptor Module
Communication Sheet**

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Pan Tilt Home Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x05\x00\x00\x00\x01\x81\x01\x06\x04\xFF
Pan Tilt Home Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x05\x00\x00\x00\x01\x81\x01\x06\x04\xFF
Pan Tilt Home Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x05\x00\x00\x00\x01\x81\x01\x06\x04\xFF
Pan Tilt Home Pan Speed 24 Tilt Speed 20	\x01\x00\x00\x05\x00\x00\x00\x01\x81\x01\x06\x04\xFF
Pan Tilt Left Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x03\xFF
Pan Tilt Left Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x14\x01\x03\xFF
Pan Tilt Left Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x01\x03\xFF
Pan Tilt Left Pan Speed 24 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x14\x01\x03\xFF
Pan Tilt Reset Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x05\x00\x00\x00\x01\x81\x01\x06\x05\xFF
Pan Tilt Reset Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x05\x00\x00\x00\x01\x81\x01\x06\x05\xFF
Pan Tilt Reset Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x05\x00\x00\x00\x01\x81\x01\x06\x05\xFF
Pan Tilt Reset Pan Speed 24 Tilt Speed 20	\x01\x00\x00\x05\x00\x00\x00\x01\x81\x01\x06\x05\xFF
Pan Tilt Right Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x02\x03\xFF
Pan Tilt Right Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x14\x02\x03\xFF
Pan Tilt Right Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x02\x03\xFF
Pan Tilt Right Pan Speed 24 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x14\x02\x03\xFF
Pan Tilt Stop Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x03\x03\xFF
Pan Tilt Stop Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x14\x03\x03\xFF
Pan Tilt Stop Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x03\x03\xFF
Pan Tilt Stop Pan Speed 24 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x14\x03\x03\xFF
Pan Tilt Up Left Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x01\xFF
Pan Tilt Up Left Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x14\x01\x01\xFF
Pan Tilt Up Left Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x01\x01\xFF
Pan Tilt Up Left Pan Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x14\x01\x01\xFF

**Global Scriptor Module
Communication Sheet**

Revision: 11/2/2021

Tilt Speed 20	
Pan Tilt Up Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x03\x01\xFF
Pan Tilt Up Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x14\x03\x01\xFF
Pan Tilt Up Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x03\x01\xFF
Pan Tilt Up Pan Speed 24 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x14\x03\x01\xFF
Pan Tilt Up Right Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x02\x01\xFF
Pan Tilt Up Right Pan Speed 1 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x14\x02\x01\xFF
Pan Tilt Up Right Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x02\x01\xFF
Pan Tilt Up Right Pan Speed 24 Tilt Speed 20	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x14\x02\x01\xFF
Power Off	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x00\x03\xFF
Power On	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x00\x02\xFF
Preset 1 Action Recall	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x02\x00\xFF
Preset 1 Action Reset	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x00\x00\xFF
Preset 1 Action Save	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01\x00\xFF
Preset 128 Action Recall	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x02\x7F\xFF
Preset 128 Action Reset	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x00\x7F\xFF
Preset 128 Action Save	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01\x7F\xFF
Shutter Down	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0A\x03\xFF
Shutter Reset	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0A\x00\xFF
Shutter Up	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0A\x02\xFF
Zoom Stop Speed 0	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x07\x00\xFF
Zoom Stop Speed 7	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x07\x00\xFF
Zoom Tele Speed 0	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x07 \xFF
Zoom Tele Speed 7	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x07' \xFF
Zoom Wide Speed 0	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x070\xFF
Zoom Wide Speed 7	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x077\xFF

Appendix D. Update Commands(Ethernet)

Auto Exposure	\x01\x10\x00\x05\x00\x00\x00\x01\x81\x09\x049\xFF
Auto Focus	\x01\x10\x00\x05\x00\x00\x00\x01\x81\x09\x048\xFF
Power	\x01\x10\x00\x05\x00\x00\x00\x01\x81\x09\x04\x00\xFF